

Principles of Economics

Week 8: The Labour Market

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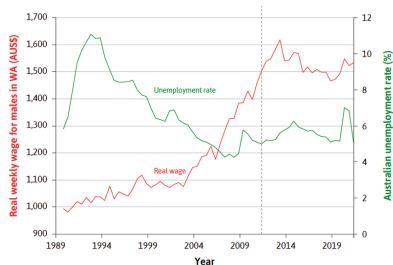
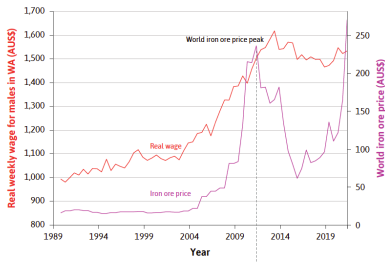


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1. Introduction

- ▶ Firms **set two key prices**:
 - ▶ Micro, Unit 6: **the wage**, with the trade-off between effort, output and costs.
 - ▶ Micro, Unit 7: **the price** of a differentiated product, via market power;
- ▶ **The labour market** (Micro, Unit 6): the conflict of interest between employer and worker.
 - ▶ A **principal-agent** model: effort is not observed;
 - ▶ As we discussed previously:
 - ▶ **Contracts cannot fully solve it**;
 - ▶ **The model predicts the equilibrium level of unemployment**.
- ▶ Today: **aggregate wages, employment and unemployment**. How are they set? Can the outcome be improved? [governments and unions]

- Real wages, final prices and the unemployment rate are related:
- Western Australia: real weekly wages for males, iron ore prices and the unemployment rate.



Plan for today's class

Introduction

Measuring unemployment

The labour market model

Wage setting

Price setting

Labour market equilibrium

Changes in aggregate demand

Changes in labour supply

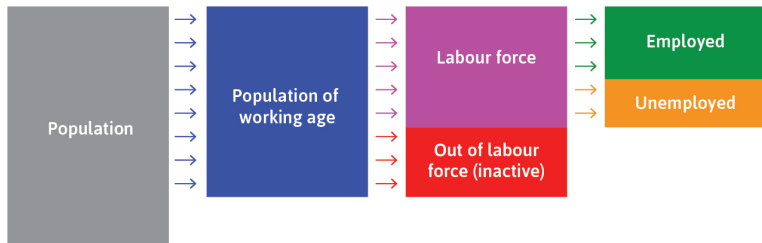
Trade unions

Labour market policies

Summary

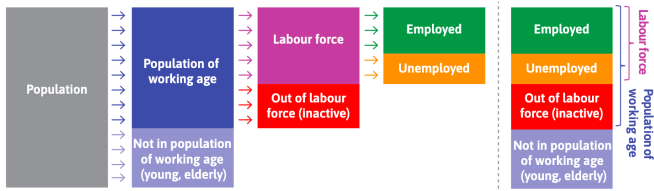
2. Measuring unemployment

- ▶ An unemployed person:
 - ▶ Is available for work;
 - ▶ Is actively searching for a job;
 - ▶ Has no paid job and is not self-employed.



Measuring Unemployment

Introduction



$$\text{Unemployment rate} = \frac{\text{Unemployed}}{\text{Labour force}}$$

$$\text{Employment rate} = \frac{\text{Employed}}{\text{Working-age population}}$$

$$\text{Participation rate} = \frac{\text{Labour force}}{\text{Working-age population}}$$

Measuring Unemployment

Introduction

► One fact per chart

Figure 2: Unemployment rate, per cent, 3 year moving average (1986-2016)

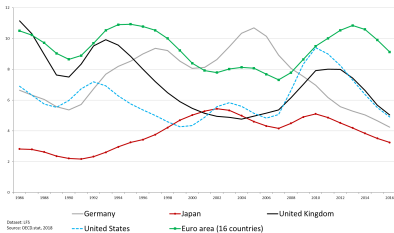
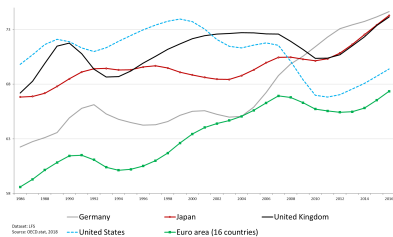
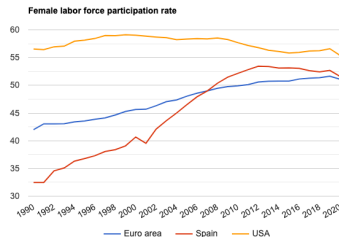
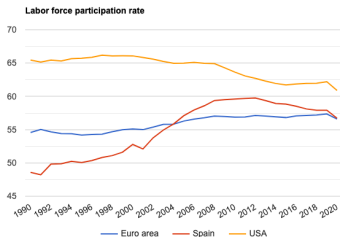


Figure 3: Employment rate, per cent, 3 year moving average (1986-2016)



- Euro area: high unemployment, low employment.
- Japan: much higher employment, low unemployment.
- UK and US: high employment, more volatile unemployment.

► One fact per chart



- Spain started low and caught up with (and passed) Europe.
- The US has always been high.
- The likely explanation: female participation.

3. The labour market model

Price and wage setting

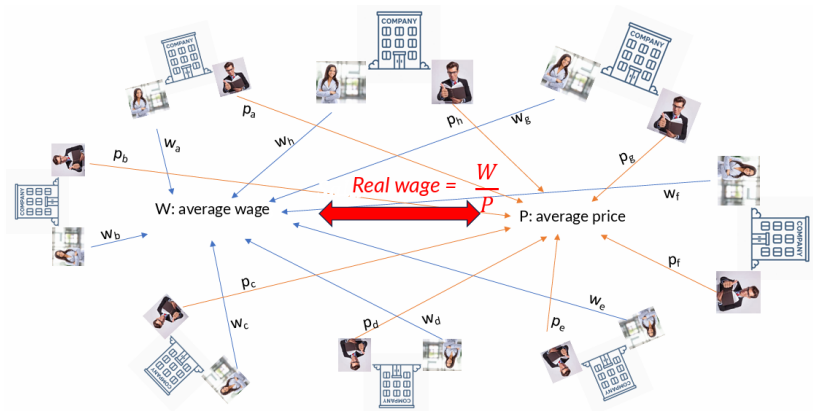
Mind map of the model

- ▶ Each firm individually sets:
 - ▶ Prices (marketing dept.):
 - ▶ Profit-maximising, given market demand $\Rightarrow$ quantity produced
 - ▶ Wages (human resources dept.):
 - ▶ High enough to buy effort $\Rightarrow$ number of workers



Price and wage setting

Mind map of the model



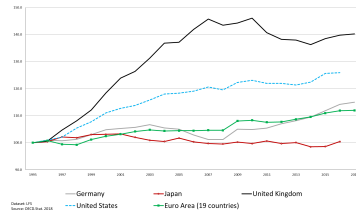
- ▶ Real wage = nominal wage divided by the price level of the consumption basket.

$$\text{Real Wage} = \frac{W}{P}$$

1. Each firm, while determining wages and prices, determines the quantity to produce and the number of workers hired.
2. Summing all of this over all firms gives total employment and the economy's real wage.

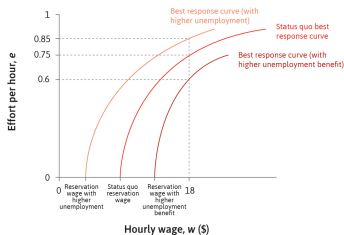
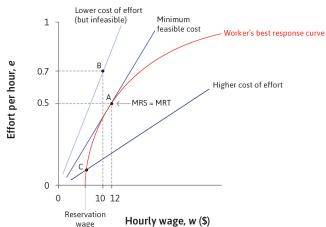
- ▶ **Real wages evolve very differently across economies:**
- ▶ **Up in the UK, and less so in the US:**
 - ▶ Nominal wages rose more than prices
- ▶ **In Japan, a downward trend:**
 - ▶ Purchasing power falls.

Figure 1: Real compensation per hour worked, deflated by CPI (1995-2017)



4. Wage setting

► Back to the labour model (Micro, Unit 6):



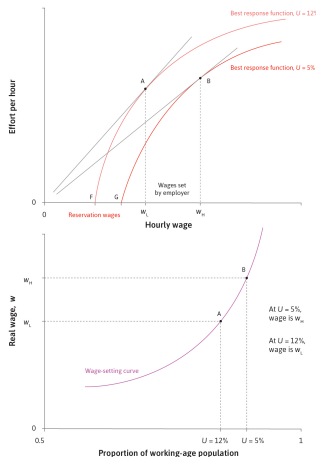
- The firm wants the steepest isocost, given the **worker's best response**.

- A change in the reservation wage shifts the best response.

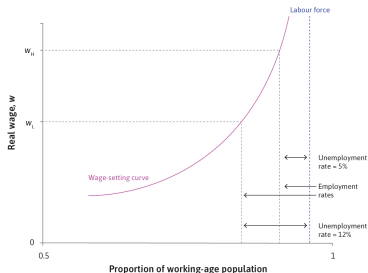
Wage-setting curve

Same labour force, more employment (less unemployment): what does the **labour model** predict?

- ▶ The best response shifts right:
 - ▶ The cost of job loss falls;
 - ▶ The reservation wage increases (F $\Rightarrow$ G).
- ▶ New equilibrium (B):
 - ▶ The real wage is higher.
- ▶ More employment, higher equilibrium wages.
- ▶ As a result, the wage-setting curve must be upward sloping:
 $\uparrow$ employment ($\downarrow$ unemployment) $\Rightarrow \uparrow$ real wage
- ▶ **This is a movement along the wage-setting curve.**



- ▶ The **wage-setting curve** summarises real wage against unemployment:
- ▶ The **wage-employment combinations** optimal for firms and workers.
- ▶ Vertical axis: Real wage
- ▶ Horizontal axis: Employment rate
- ▶ **Labour force**: those actively seeking work.
- ▶ **Unemployment rate**: the horizontal distance between labour force and employment.



Wage determination

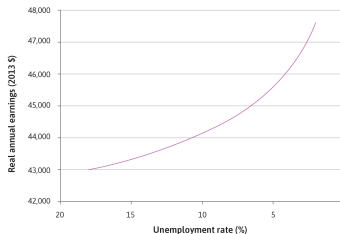
Wage-setting curve

- From the firm to the whole economy



Wage-setting curve

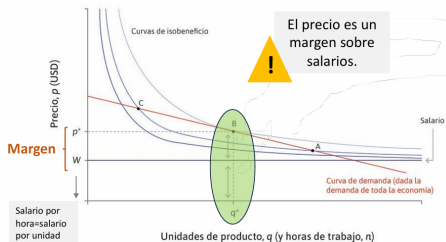
- ▶ The **wage-setting curve** can be estimated from administrative data on wages and unemployment:
- ▶ Using data from the United States:
- ▶ The positive slope is empirically validated!
- ▶ Careful with the axis: unemployment grows to the left, which is employment growing to the right, as before.



5. Price setting

Profit-maximising price

- ▶ The optimal price is the tangency between demand and the iso-profit curve (Micro, Unit 7)
- ▶ That fixes output, and so the number of workers needed.
- ▶ Two simplifying assumptions:
 1. Labour is the only factor of production.
 2. A worker produces one unit per hour.
- ▶ So, what is the marginal cost?
 - ▶ The hourly wage.



Profit-maximising price

- ▶ Recall the markup formula:

$$\frac{P_i - w_i}{P_i} = \frac{1}{\epsilon_i} \iff \frac{w_i}{P_i} = 1 - \frac{1}{\epsilon_i}$$

where P_i , w_i , ϵ_i are, respectively, prices, wages and the elasticity of the firm's demand.

- ▶ And if each worker produces λ_i , that is, their **productivity** is λ_i .
- ▶ Then the MC is $\frac{w_i}{\lambda_i}$.

$$\frac{P_i - \frac{w_i}{\lambda_i}}{P_i} = \frac{1}{\epsilon_i} \iff \frac{w_i}{P_i} = \lambda_i \left(1 - \frac{1}{\epsilon_i}\right) = \lambda_i (1 - \text{markup}_i)$$

- ▶ The markup is the share of the price that goes to profit; the rest is the wage.
- ▶ If every firm does this, the economy's real wage is:

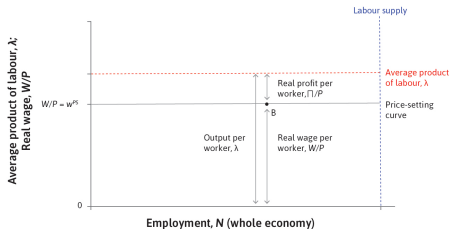
$$\frac{w_i}{P_i} = \lambda_i (1 - \text{markup}_i) \iff \frac{w}{P} = \lambda (1 - \text{markup})$$

- The real wage in the economy:

$$\frac{w_i}{P_i} = \lambda_i(1 - \text{markup}_i) \iff \frac{w}{P} = \lambda(1 - \text{markup})$$

What happens if?

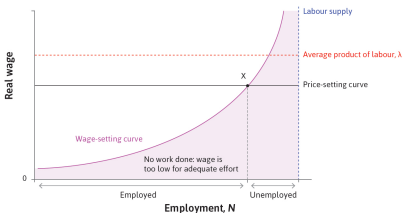
- Competition falls (market power rises).
- Productivity in the economy increases.
- Prices rise and wages do not.



6. Labour market equilibrium

Determination of the equilibrium unemployment level

- ▶ Point X is a **(Nash) equilibrium**. Why?
- ▶ Everyone is doing their best, given what the others do.
- ▶ Firms offer the **optimal wage** (the lowest one that ensures optimal effort);
- ▶ Workers exert their **optimal effort**.
- ▶ Firms set **optimal prices** (with their markup)
- ▶ **There is unemployment.**
WHY?
- ▶ At those prices and wages firms already produce their optimum: they need no more workers.
- ▶ And with lower wages?
- ▶ Not worth it: effort would fall too far.



- ▶ Unemployment, the **excess supply of labour**, is a feature of the equilibrium, not a failure of it:
- ▶ With no unemployment, **job loss costs nothing and effort disappears**;
- ▶ It is the possibility of unemployment that motivates [Micro, Unit 6].
- ▶ Firms raise wages to buy effort...
- ▶ ...but that makes production dearer, so they produce less.
- ▶ Less production, fewer workers...
- ▶ ...unemployment appears, exactly the element needed to incentivise effort.

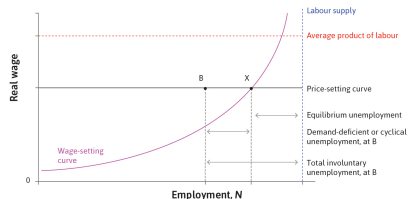
7. Changes in aggregate demand

Involuntary unemployment

- ▶ **Aggregate demand:** the demand for every good and service in the economy.
- ▶ If unemployment follows **firms' demand for workers**:
 - ▶ it **depends on firms' demand for products and services**, aggregate demand.
- ▶ Unemployment caused by a fall in aggregate demand: **demand-deficient unemployment**.
- ▶ But **high unemployment** reduces aggregate demand:
 - ▶ It cuts **purchasing power** and consumption;
 - ▶ Which **compresses demand further**, and unemployment with it.

- ▶ **Low aggregate demand** cuts employment: what happens next?
- ▶ The economy moves from X to B [along the price-setting curve]
- ▶ But B is not an equilibrium:
 1. Firms can **cut wages** without losing effort;
 2. Lower production costs allow **price reductions**;
 3. Lower prices stimulate demand and **output rises**;
 4. To produce more, firms **hire more workers**.
 5. Employment **returns to equilibrium (X)**.

- ▶ So imbalances would be self-correcting.



Is this self-correction mechanism realistic?

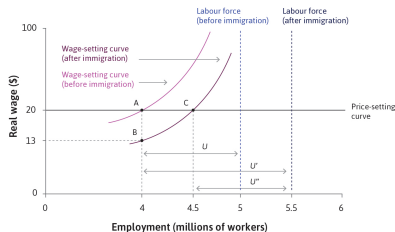
- ▶ Real economies **do not work so well**:
- ▶ Workers resist nominal wage cuts:
 - ▶ Less motivation, and strikes;
 - ▶ And often it is not legally possible [unions, later today]
- ▶ With lower wages, people spend less and **aggregate demand falls further**;
- ▶ Falling prices **make consumers postpone purchases**, waiting for cheaper still:
 - ▶ This could lead to deflation (which can be dangerous, as we will see)
- ▶ If any of these bites, **the economy stays trapped below equilibrium employment**:
 - ▶ In fact, employment may fall even further!

- ▶ One way out: **government intervention**.
- ▶ **Expand aggregate demand** with **fiscal or monetary policy**:
 - ▶ Increase public consumption;
 - ▶ Reduce taxes;
 - ▶ Devalue the currency to stimulate exports.
- ▶ Equilibrium output rises, and:
 - ▶ Increase employment from B to X;
 - ▶ Increase profits without lowering wages.

8. Changes in labour supply

The role of labour supply

- ▶ **Labour supply** also moves the equilibrium:
- ▶ Say immigration raises labour supply:
 - ▶ This shifts labour supply to the right;
 - ▶ And **increases unemployment** ($U \rightarrow U'$)
- ▶ With **more people looking for work**:
 - ▶ Increase in labour rents;
 - ▶ Lower cost of effort;
 - ▶ The wage-setting curve shifts to the right.
- ▶ Now, the **initial employment level is not optimal (B)**:
- ▶ Firms hire more workers;
- ▶ Employment increases and **unemployment decreases** from ($U' \rightarrow U''$).



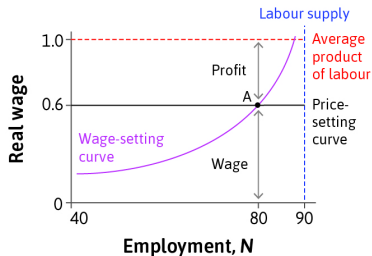
9. Trade unions

The role of trade unions

Inequality in the division of output

- ▶ The labour market **divides the economy's output** among:
 - ▶ Employed workers, the unemployed and firm owners.

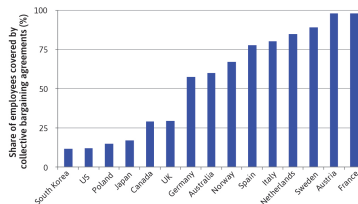
- ▶ Workers may be unhappy with that split, above all if it is **very unequal**.



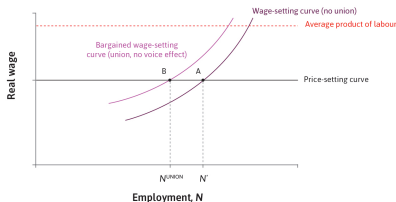
The role of trade unions

Unions and wages

- ▶ **Trade unions** represent employees and push for better wages and conditions.
- ▶ **Their weight** depends on the **share of employees covered**:
- ▶ If unions are strong (high bargaining power):
 - ▶ Wages are not set unilaterally by the employer;
 - ▶ They are bargained between union and firm.



- ▶ The **bargained wage** can sit **above the wage-setting curve**:
 - ▶ The threat of collective action, such as a strike, gets them there.
- ▶ **How would this change the equilibrium, compared to a case with no unions?**
 - ▶ A higher (reservation) wage raises firms' costs;
 - ▶ The wage-setting curve shifts left;
 - ▶ And, in equilibrium, **leads to lower employment**.
- ▶ **A potentially adverse effect of unions:**
 - ▶ Wages do not change;
 - ▶ Employment decreases;
 - ▶ Profits probably fall.

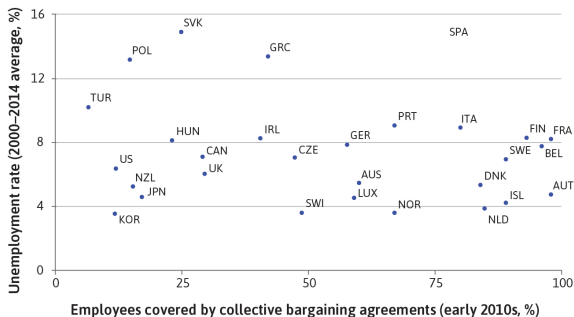


The role of trade unions

Unions and wages

- Is there a **positive** relationship between union power and unemployment?

- The data do not seem to suggest so!

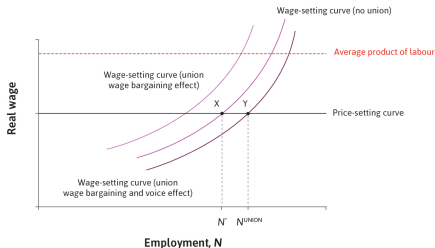


- One explanation: the **union voice effect**.

The role of trade unions

Unions and wages

- ▶ Giving employees a voice can buy **more participation and effort at the same wage**.
- ▶ Then the bargained wage curve shifts right;
- ▶ And **equilibrium employment rises**.
- ▶ The overall effect of unions on employment is **ambiguous**.



10. Labour market policies

Shifts in price or wage setting, or in the labour supply curve

- ▶ **Policies move the equilibrium** by shifting the curves:
 - ▶ Changes in the **price-setting curve**:
 - ▶ Education and training raise productivity (price-setting curve up)
 - ▶ A wage subsidy cuts costs and prices (price-setting curve up)
 - ▶ Changes in the **wage-setting curve**:
 - ▶ Lower unemployment benefits lower the reservation wage (wage-setting curve down)
 - ▶ Changes in **labour supply**:
 - ▶ More open immigration raises labour supply (supply curve right)
 - ▶ Childcare raises female participation (supply curve right)
 - ▶ (and labour supply moves the wage-setting curve and the equilibrium, as we saw)

11. Summary

Section by section (Macro, Unit 1)

1. **Measuring unemployment** → labour force, employment, unemployment and participation rates; the Euro area and Japan are two ends of the same map.
2. **The labour market model** → each firm sets a price and a wage; adding up gives employment and the real wage W/P .
3. **Wage setting** → more employment, less to lose from dismissal, higher wage: the wage-setting curve slopes upward.
4. **Price setting** → $w/P = \lambda(1 - \text{markup})$: productivity and competition fix the real wage firms can pay.
5. **Labour market equilibrium** → X is a Nash equilibrium *with* unemployment: without it there would be no effort.
6. **Changes in aggregate demand** → demand-deficient unemployment; the self-correction is slow, and government can shorten it.
7. **Changes in labour supply** → immigration raises unemployment first and employment afterwards.
8. **Trade unions and policies** → bargaining lowers employment, the voice effect raises it: the net effect is ambiguous, and the data show no clear relationship.

Key concepts: labour force, unemployment and participation rates, real wage, wage-setting curve, price-setting curve, markup, productivity, Nash equilibrium with unemployment, demand-deficient unemployment, aggregate demand, trade union, voice effect.